

VISHAL SANKAR RAM

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EDUCATION

University of Southern California (Master of Science in Computer Science, GPA: 3.85/4.0) Aug 2024 - Present

- **Relevant Coursework:** CSCI 585 Database Systems, CSCI 570 Analysis of Algorithms, CSCI 566 Deep Learning and Its Applications, CSCI 544 Applied Natural Language Processing.

National Institute of Technology, Tiruchirappalli, GPA: 8.84/10

Bachelor of Technology in Instrumentation and Control Engineering, Minors – Computer Science Jul 2018 - May 2022

- **Relevant Coursework:** Basics of Programming, Data Structures & Algorithms, Operating Systems, Pattern Recognition, Database Management Systems, Web Technology, Data Communications and Networks, Software Engineering

SKILLS

- **Technical Skills & Frameworks:** Python, Java, C, C++, JavaScript, Dart, React, Node, Express, Flutter, REST API, PostgreSQL, MongoDB, Firebase, Redis, Neo4j, DynamoDB, AWS, GCP, Docker, Kubernetes, CI/CD, Machine Learning, Reinforcement Learning, GenAI, Apache Spark, JUnit, Git, JIRA, Agile, Jenkins, Kubeflow, Flask, Redux, XML, Maven, Snowflake, Kafka, Docker, Kubernetes, REST APIs, JWT, TensorFlow, PyTorch, Scikit-learn, MapReduce, NovaLSM, Selenium, Valgrind, HTML, NoSQL, MySQL.

EXPERIENCE

CPS-Vida Lab, USC (Researcher), Los Angeles Dec 2024 – Present

- Conducted an in-depth literature review on advanced Reinforcement Learning (RL) algorithms, including Rainbow DQN and Conservative Q-Learning, to identify cutting-edge techniques for real-world problem-solving.
- Collaborated with a team to develop and test RL-based solutions for humanoid animal models with real-world applications.

Samsung R&D Institute India, Bangalore (Senior Engineer), India Aug 2022 – Aug 2024

Automatic Speech Recognition – enIN Modeling Team

- Boosted Named Entity accuracy by 5% through Conformer and Transformer models, facilitating Bixby's delivery of precise responses
- Increased contextual utterance accuracy by 6% by creating a WFST based Shallow Fusion Language Model to boost precision.
- Enhanced ASR performance by 10% for long-form audio by optimizing the segmented audio stitching algorithm for greater fluency.

Text-to-Speech Team

- Developed a robust Test-Driven TTS front-end module in C for Indic languages, decreasing memory leaks, enhancing system stability
- Streamlined the Tokenizer, Text-Normalizer, driving a 0.25 boost in MOS ratings, enhancing speech clarity for Galaxy AI applications
- Implemented a fully automated Jenkins pipeline using Bash scripts, significantly accelerating the linguist team's review process and enabling real-time issue tracking and updates, resulting in a 30% reduction in turnaround time and improved team efficiency.

Name-Entity Replacer

- Engineered a high-performance Named Entity Replacer using Java & Lucene with TF-IDF, enhancing ASR output customization.
- Boosted entity replacement precision using advanced heuristics and phonetic hashing, significantly improving information retrieval.
- Analyzed 15GB of real-time text data monthly, implementing optimizations that promotes system performance by 10% and efficiency

Inverse Text Normalization Team

- Refactored and optimized legacy Java code in a Linux environment, improving ASR output readability across Call and App domains.
- Automated regression testing using Gradle, boosting consistency, performance, and accelerating deployments improving reliability.

American Express (India) Private Limited (Software Engineering Intern), Remote May 2021 – Jul 2021

- Developed a React.js web application to manage and secure confidential employee data using a Node.js and SQL backend.
- Programmed secure RESTful API authentication using OAuth 2.0 & JWT to ensure user identity verification & data access control, while optimizing ID-based indexing to enhance data handling & retrieval efficiency for improved query performance.
- Improved performance by 15% through Lazy Loading and code-splitting, reducing client-side load and enhancing user experience.

PROJECTS

Database System Exploration & Performance – USC, 2024-2025

- Analyzed FoundationDB operations using Java and C APIs, evaluating key-value storage efficiency through advanced range queries, parallel processing, and performance benchmarking across streaming modes with datasets of 10,000+ key-value pairs.
- Modeled graph data with JanusGraph, leveraging BerkeleyDB and FoundationDB backends; built and analyzed AirRoutes schema with 4,000 vertices and 60,000 edges to evaluate load times and storage efficiency.

Video Based Vehicle Detection and Counting System for Route Planning – NIT Trichy, 2022

- Applied advanced background subtraction techniques to accurately identify and isolate vehicles from complex backgrounds in various traffic conditions and leveraged computer vision algorithms to enable real-time vehicle detection and tracking
- Implemented centroid-based tracking for real-time vehicle movement analysis, enhancing performance in dynamic environments, and modified Dijkstra's Algorithm for optimizing travel routes, incorporating real-time traffic data to reduce travel time.

Connect-IT – NIT Trichy, 2019

- Developed social networking software using Android Studio and Java, with a user-friendly interface and features, integrating database for managing complex relationships and enabling data-driven insights through advanced user behavior analytics.
- Integrated Firebase for real-time user action, data storage and synchronization, and Firebase Auth for secure user authentication.